Paleomagnetism of Luzon

M. FULLER, R. MCCABE, AND I. S. WILLIAMS

Department of Geological Sciences, University of California, Santa Barbara, California 93106

J. ALMASCO, R. Y. ENCINA, AND A. S. ZANORIA

Philippine Bureau of Mines, Manila, Philippines

J. A. WOLFE

Pan-Asian Technical Services, Manila, Philippines

Paleomagnetic studies of 50 Cenozoic sites in the northern Philippines are reported and used to constrain tectonic models of the evolution of the region. Plio-Pleistocene results demonstrate that paleomagnetically distinguishable rotations of the regions have not taken place in the past 5 m.y. Early and mid-Miocene results reveal counterclockwise rotation of approximately 20°, but give no evidence of detectable N/S translation. Oligocene results from the Cordillera of Luzon near Baguio give similar rotations to those found in the early and mid-Miocene formations. Eocene sites from the Zambales ophiolite and overlying sediments indicate stronger rotations and appear to have been formed at equatorial latitudes. A tectonic model for the past 17 m.y. is proposed which is constrained by the counterclockwise rotation of early- and mid-Miocene sites. The model invokes the combined action of the northwestward motion of the Philippine Sea Plate and the pinning of the northern Philippines at its southern margin by collision with the Calamian continental fragment. The continued advance of the leading edge of the Philippine Sea Plate causes counterclockwise rotation of Luzon and the Manila trench, until the northern end of the trench becomes pinned by collision with Taiwan.

1. Introduction

The Philippine Islands lie off the Asian landmass between the South China and Philippine Seas, part of the string of islands from the Kuriles in the north to Indonesia in the south. At the latitude of the Philippines, more than two thousand kilometers separate the Pacific plate from the Asian landmass. This tectonically active region is the site of formation of new ocean crust, of subduction, and of major transform faulting. In addition to these fundamental plate interactions, new continental crust is being formed. The processes forming these new landmasses may well be similar to those that operated, in more remote geological time, to produce the continents we now see. An understanding of the geology of the Philippines and of the tectonically active region in which they lie may therefore be of some general interest in explaining the growth of continents.

The geology of the Philippines records the complicated history of a mobile region caught between two major plates. The first paper on the geology of the Philippines was written at the very end of the last century [*Becker*, 1899]. Since that time, many workers have attempted regional descriptions [e.g., Corby, 1951]. During the 1960's, the Philippine Bureau of Mines compiled much of this work and published the *Geologic Map of the Philippines* [*Philippine Bureau of Mines*, 1964], at a scale of 1 : 1,000,000. In the late 1960's, *Gervasio* [1967] published a tectonic synthesis of the Philippines. With the advent of plate tectonics and in particular the development of ideas on the origin of marginal basins [*Karig*, 1971], the geology of the area in which the Philippines lie became a matter of considerable general interest; papers appeared interpreting the Philippines in terms of plate tectonic concepts [e.g., *Karig*, 1973] in such a complicated mobile region, an essential aid to our understanding is a paleomagnetic framework against which tectonic models can be tested.

Hsu [1972] carried out the first paleomagnetic study in Luzon, analyzing a large collection of samples. His conclusions were that Luzon had suffered continuous counterclockwise rotation throughout the Tertiary and had moved from south to north at a rate of 10 cm per year.

Copyright American Geophysical Union

This study may well have established major elements of the paleomagnetic framework of Luzon. However, Hsu did encounter serious experimental difficulties. Moreover, the study was carried out before the advent of cryogenic magnetometers and before recent improvements in analysis of paleomagnetic data. It is therefore important that the study be refined and extended. More recently, *De Boer et al.* [1980] reported a study of Plio-Pleistocene volcanics in central Luzon, suggesting that the region had suffered counterclockwise rotation of about 35° in the past 4 million years. This rotation rate is remarkable, even for so tectonically active a region as the Philippines, and is not supported by the earlier study of *Hsu* [1972]. An excellent review of paleomagnetic studies in Southeast Asia, including an analysis of the tectonic implications of the seafloor magnetic anomalies in the area, has been given by *Jarrard and Sasajima* [1980].

The paleomagnetic study described in this paper is designed to extend and refine the earlier studies and, in particular, to answer the following two questions. (1) What were the tectonic rotations suffered by Luzon during the Tertiary? (2) Has Luzon acted as a single tectonic unit during this time, or can separate histories be determined for different units within the island?

Clearly, the answers to these questions have important bearing on the geological history of the Philippines themselves and of the surrounding regions.

2. Paleomagnetic Techniques

Standard paleomagnetic techniques were used throughout this study. The samples were mostly collected with a portable drill, although, at some sites, it was more convenient to collect hand samples, which were later drilled at the laboratory. The measurements of remanent magnetization were made for the most part with an SCT horizontal access magnetometer. Particularly strongly magnetized samples were measured with a Schonstedt spinner. The magnetometers have dedicated North Star minicomputers, which carry out the data reduction. Six measurements are made of each sample with the SCT magnetometer and Fisher statistics [*Fisher,* 1953] used to assess the measurement of the individual sample.

Alternating field and thermal demagnetization were used to isolate the primary natural remanent magnetization (NRM) from secondary magnetizations. Static AF demagnetization was used with a coil capable of giving a 1000 Oe (79.58×10^3 At/m) H field. The coil was placed in a three-stage μ-metal shield and driven at resonance by a Bellman tunable AC power supply. No single level of AF demagnetization was chosen on the basis of pilot studies, but, rather, each sample was carried through a sequence of demagnetizations. Preferred directions were chosen by using Zijderveld diagrams [*Zijderveld,* 1967]. Nonconvergence to the origin on Zijderveld diagrams and vector differences, which remained distinct from the demagnetized directions, were used as criteria to exclude results. The North Star minicomputer drives a Houston plotter, so that the Zijderveld diagrams and equal area plots of unit vectors may be produced and studied during demagnetization.

Thermal demagnetization was carried out with two systems. One, which relied upon field cancelation by a coil array, gave cancelation to a part in 10^3 of the field. The other system, which had a three-stage μ-metal shield, gave cancelation to a part in 10^4. The results of thermal demagnetization were analyzed in a similar manner to that used with the AF demagnetization.

The analysis of these samples from the Philippines was unusually difficult. Part of this difficulty arose from weathering of some of the samples, which had generated stable secondary magnetizations not readily separated from the primary NRM. About 20% of the sites had to be rejected because the NRM after demagnetization was still erratic. Those sites which are reported generally had a α_{95} of less than 10.

Tectonic corrections were applied in the standard manner, but in no case did we have sufficient geological field data to apply a correction for plunging folds. The tectonic corrections applied to dikes was that of the bedding of the oldest sediments not intruded by the dike. The choice of the tectonic correction in the Zambales ophiolite presented special problems and is discussed below.

Fisher [1953] statistics were used throughout. The final analysis is given in terms of directions of magnetization.

3. Geological Setting of Sampling Sites

The principal collections were made in the vicinity of Baguio in Benguet Province and in Zambales and Pangasinan. Subsidiary collections were made in the various provinces shown in the map of the northern Philippines (Figure 1), such as Rizal, Albay, and Marinduque. The samples range in age from Eocene to Pleistocene. Although they do not give total coverage across Luzon, they do represent different major structural regions of the island. For example, the two major collections permit comparison of the tectonic history of the Central Cordillera represented by Baguio and the Zambales ophiolite and associated formations.

The geology of the Baguio region has recently been reviewed by *Balce et al.* [1980]. As they note, the tectonic development of the Luzon Central Cordillera is related to the north-south subduction zones lying offshore on either side of the Philippines. To the west lies the east dipping Manila subduction zone. To the east there is the west dipping subduction zone of the Philippine trench. Baguio itself is located at the southern end of the Central Cordillera, where the Cordillera is terminated by the Philippine fault. The area is of great economic interest, providing major contributions to the gold and copper production of the Philippines. As a consequence, its geology has received considerable attention.

The Baguio region affords access to important Tertiary

Copyright American Geophysical Union

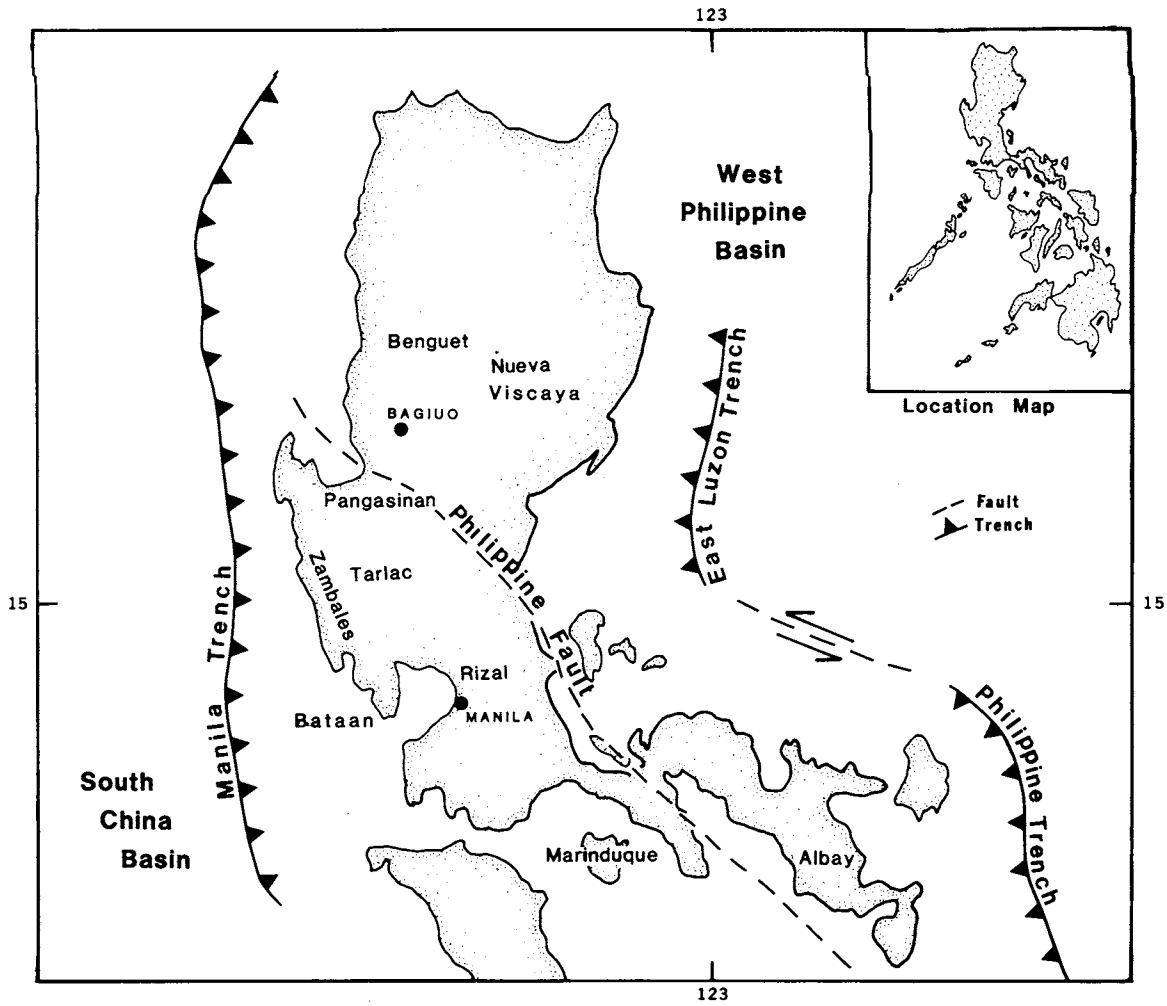


Fig. 1. Sketch map of the northern Philippines showing the major structural features and the provinces from which paleomagnetic samples were taken.

sedimentary sections, as well as to flows and intrusions of the same age. Exposures are good due to the rapid uplift and associated high relief in the region. Figure 2 gives a simplified stratigraphic column and Figure 3 illustrates the various sites. Immediately to the south of Baguio, the Kennon Road follows the Bued River through a region of rugged relief in which Oligocene to Pliocene sediments are exposed. There are also minor intrusions and flows. This formed one of the main collecting sections. To the west of Baguio, the Asin, Naguillian Roads, and the new Marcos Highway give somewhat comparable sections through upper Miocene and Pliocene sediments. Some 30 sites are reported from the region.

Zambales lies to the southwest of Baguio and is separated from the Central Cordillera by the Central Basin. The region is dominated by the Zambales range composed of the ultrabasics and gabbros of the Zambales ophiolite complex [e.g., *Hawkins and Evans,* this volume]. Our collection sites are illustrated in Figure 4 and the stratigraphic column is given in Figure 5. The main collections were from the gabbros at the Acoje and Coto mines. In addition, we sampled sills, gabbros, and a quartz diorite to the north near Sual. At Barlo, we collected Oligocene sediments. On the east side of the range, sills in the Camelling River, and the Eocene Aksitero formation, which lies conformably above the pillow basalts of the ophiolite complex [*Schweller and Karig,* 1982]. Above that, the Miocene Moriones formation was also sampled.

4. Paleomagnetic Results

The paleomagnetic results are presented in Tables 1–5 and Figures 6–10. They are discussed in order of increasing age from Plio-Pleistocene to Eocene. In each of the figures, the small stereonets show the directions of mag-

Copyright American Geophysical Union

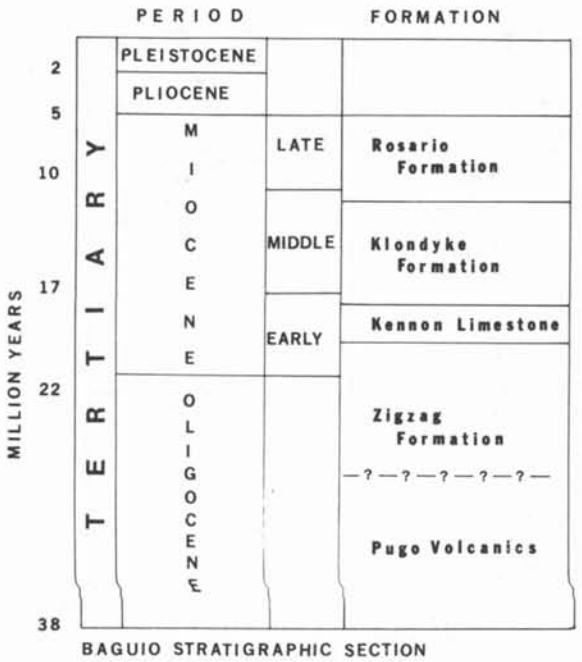


Fig. 2. Stratigraphic section for the Baguio area (Benguet province); after *Balce et al.* [1980].

netization of the individual samples from each site. The large stereonets give the site mean directions with their associated α_{95} ovals.

4.1. *Pliocene-Pleistocene Results*

There are 14 sites in Pliocene-Pleistocene age formations from five provinces of Luzon. The results from Benguet, Tarlac, and Rizal have declinations indistinguishable from zero; however, their inclinations are consistently shallower than that predicted by the geocentric axial dipole. The direction from the Albay site is marginally distinguishable at the 95% confidence level. However, these results come from volcanic units that cool rapidly, giving spot readings of the field; they do not, therefore, average secular variation. The Fisher mean of all the Bataan sites yields a declination of 354.7, indistinguishable from zero at the 95% confidence level.

These results suggest that the mean declination for Pliocene-Pleistocene formations on Luzon is not displaced from zero. The mean inclination is 17.1 and is statistically distinguishable from that of the axial geocentric dipole. The mean virtual geomagnetic pole (VGP) estimate from all the Luzon sites is located at N83.8 W26.9 with an α_{95} of 3.6.

4.2 *Late Miocene*

The late Miocene results come from Benguet, Marinduque, and Albay provinces. The sites on Marinduque are indistinguishable from the geocentric axial dipole direction. Although two sites near Legaspi, in Albay Province, are distinguishable from the dipole field, the mean of the three sites is not. Thus, the departures could be due to secular variation.

The four sites from the Baguio City region of Benguet Province give inconsistent results. One site has a declination of 009. The two reversed sites have equivalent normal declinations of 018 and 033. Site 78-10 is an andesite flow and has a single component of magnetization. Site 79-26 is a sediment from the Rosario formation a few hundred meters stratigraphically beneath the flow. Site 81-29 is a flow in the Santo Tomas area, which, like the other flow, is paleomagnetically well behaved. Site 79-25, which has a declination of 348, is stratigraphically lower in the Rosario section and has a direction close to the present field before tectonic correction. It therefore appears that the most reliable sites from the Baguio region

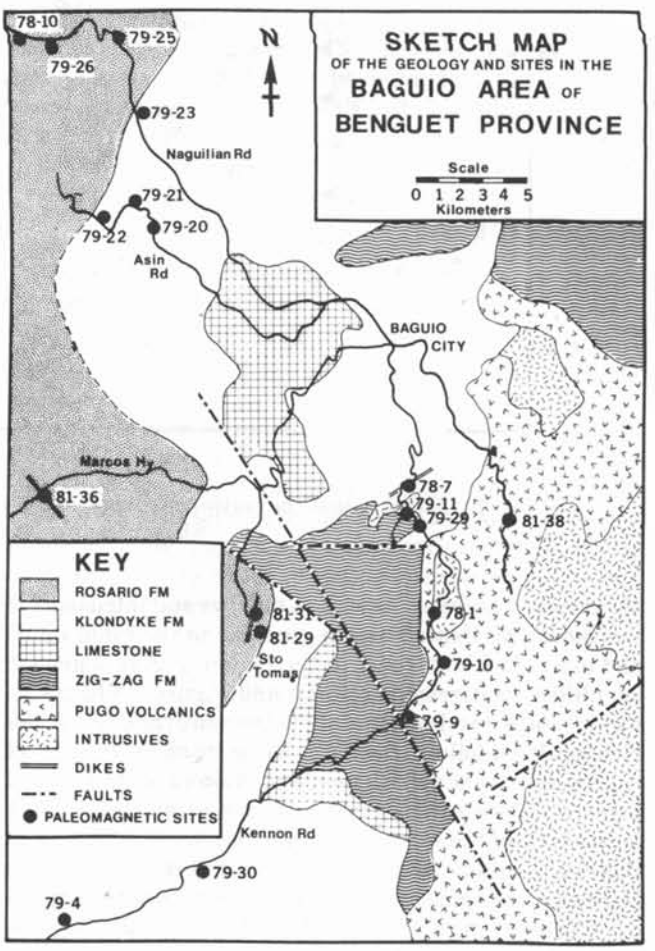


Fig. 3. Sketch map of the geology of the area around Baguio city in Benguet province. The geology is simplified and is based upon that shown by *Balce et al.* [1980], *Pena* [1970], and the *Philippine Bureau of Mines* [1964].

Copyright American Geophysical Union

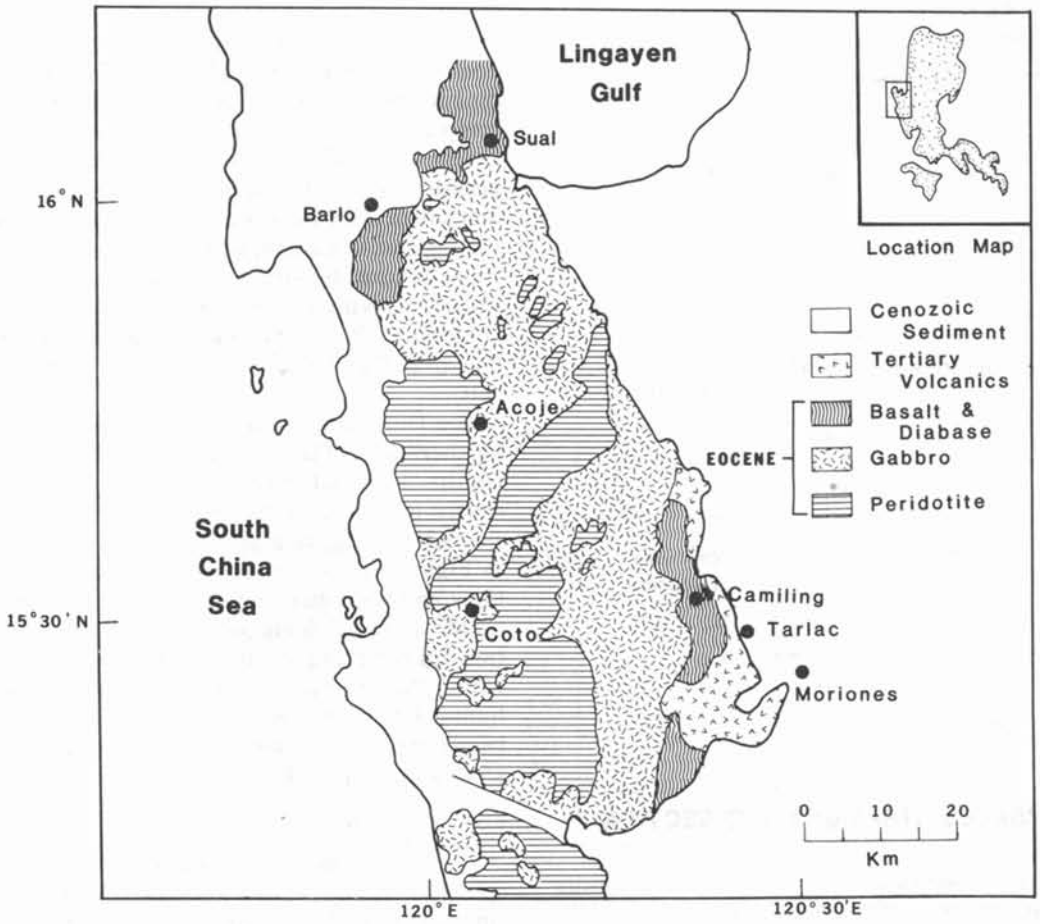


Fig. 4. Sketch map of the geology of the Zambales Ophiolite, in Zambales and Pangasinan provinces, based upon that by *Hawkins and Evans* [this volume].

of late Miocene age all have easterly declinations. However, the mean direction of these three sites is not statistically distinguishable from the geocentric axial dipole field direction.

The late Miocene results from the Baguio region present something of a puzzle, which is not likely to be resolved until more extensive collections are made. Taken at face value, they suggest that the region may have recorded a different declination from those found in Marinduque and Albay. However, until the result is demonstrated by more extensive collections, one should be cautious in its use.

4.3. *Early and Mid-Miocene Results*

The early and mid-Miocene results give consistently westerly declinations. All of the sites from Benguet, Marinduque, and Tarlac show this westerly declination, although the determinations from Tarlac and Marinduque do not provide as high quality data as those from Benguet.

The results from Baguio in Benguet come from sediments, a dyke, and a stock. Before tectonic correction, the Klondyke sites have a direction close to, but distinguishable from, the present field. The reversed sites from the Zig-Zag formation and from the dyke are, however, antipodal to the tectonically corrected Klondyke results. Since the reversed sites are clearly not remagnetized in the present field, their antipodal relation to the Klondyke sites gives increased confidence in the Klondyke results. The mean direction for the Baguio results is D. 344.2 I. + 30 with an α_{95} of 5.2.

The Miocene results present one of the clearest patterns we have found. Over a considerable region a variety of rock types define a westerly declination. The results from Baguio are particularly convincing. The mean inclination of all sites is indistinguishable from the geocentric axial dipole field.

4.4. *Oligocene Results*

As Figure 9 and Table 4 reveal, the Oligocene results are much sparser than those for other periods. The results come from Benguet, Nueva Vizcaya, and Pangasinan. The

Copyright American Geophysical Union

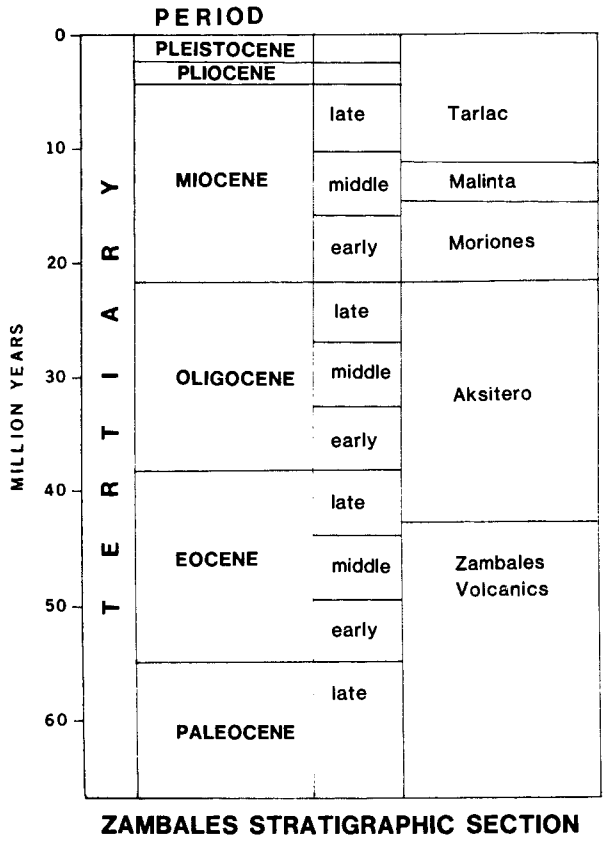


Fig. 5. Stratigraphic section for the Zambales area (Zambales and Pangasinan provinces); after *Schweller and Karig* [1982].

latter province is in the northern part of the region dominated by the Zambales ophiolite.

The results from Benguet and Nueva Vizcaya (immediately to the east) reveal declinations of between 320° and 340°, which are indistinguishable from the early and mid-Miocene westerly declinations. The Pugo volcanics series, from which the results from near Baguio City come, has been subjected to low grade metamorphism. However, the magnetization does not seem to have been affected; a single component is seen, and it is unlikely that the magnetization would have been completely reset by such low temperature regional metamorphism. In any case, the metamorphism must have occurred prior to the early Miocene; the Zig-Zag formation, which lies unconformably upon the Pugo, is unmetamorphosed. It therefore appears that the Pugo is carrying westerly declinations acquired during the Oligocene. The single result from an unmetamorphosed flow from Caraballo in Nueva Vizcaya gave a similar result.

The Oligocene sediments from Pangasinan lie on the northern part of the Zambales ophiolite and give puzzling results. They will have to be re-sampled. The sediments consist of buff and very light-colored siltstones. The light-colored facies gave a good grouping at 300 Oe as shown in Figure 9. The darker siltstones had negative inclinations and were distributed in declination from about 030 to 080 after AF demagnetization. Thermal demagnetization did not bring about a better grouping. Taken at face value, the appropriate recorded direction is that of the light-colored siltstones. However, the inconsistency between the two sediment types suggests that additional analysis is required.

4.5. *Eocene Results*

The Eocene results come exclusively from the Zambales ophiolite and associated formations in Zambales, Pangasinan, and Tarlac provinces. Most of the igneous units sampled gave paleomagnetically well-behaved directions, but there is considerable difficulty in assigning appropriate tectonic corrections. The sediments, which are not such good paleomagnetic recorders, do provide some control for tectonic corrections.

On the beach at Sual, there are tabular intrusives that

TABLE 1. Pliocene-Pleistocene Paleomagnetic Results

Province	Site	Formation	Location		N	Demagnetization*	Inc.	Dec.	α_{95}	K	VGP Latitude	VGP Longitude
Benguet	81-31	dike	N16.3	E120.5	7	400/450	14.0	359.5	8.8	48	N80.8	W56.4
	81-36	dike	N16.3	E120.5	4	—/450	−18.3	170.2	10.1	24	S78.2	W175.8
Tarlac	81-51	Bamban	N15.3	E120.6	7	200/450	26.1	1.0	7.4	68	N88.2	W91.7
	81-52	Bamban	N15.3	E120.6	9	200/550	−16.3	176.3	9.8	28	S82.1	W148.4
Rizal	81-61	Antipolo	N14.5	E121.1	6	900/400	−27.9	175.4	5.1	175	S85.5	W144.2
	81-99	Guadalupe	N14.6	E121.0	9	100/400	−12.9	182.1	7.7	45	S81.8	E106.6
	81-100	Guadalupe	N14.6	E121.2	8	150/450	−23.8	183.4	10.7	24	S86.1	E63.3
	81-101	Guadalupe	N14.6	E121.2	11	—/450	−18.2	183.8	8.1	36	S83.6	E85.6
Albay	81-203	flow	N13.5	E123.6	9	300/550	19.1	352.6	4.8	117	N81.9	E7.5
Bataan	81-110	Mariveles†	N14.4	E120.4	6	300/500	3.3	4.2	6.7	190	N76.6	W78.0
	81-111	Mariveles	N14.4	E120.4	11	300/550	−22.3	171.1	1.9	518	S80.9	W166.5
	81-112	Mariveles	N14.4	E120.4	9	300/550	−19.7	177.5	3.2	260	S85.1	E150.6
	81-113	Mariveles	N14.4	E120.4	5	300/—	−12.3	175.6	4.1	346	S80.7	E147.7
	81-114	Mariveles	N14.4	E120.4	6	300/—	−4.6	165.2	7.4	83	S71.0	E172.1

N is the number of individually oriented paleomagnetic samples; Inc., inclination; Dec., declination; VGP, virtual geomagnetic pole.
*Demagnetization values are listed for first A.F. demagnetization in Oersteds, then thermal demagnetization in degrees Celsius.
†K-Ar age of 3.26 ± 0.6 m.y. [*Wolfe*, 1981].

Copyright American Geophysical Union

TABLE 2. Late Miocene Paleomagnetic Results

Province	Site	Formation	Location		N	Demagnetization*	Inc.	Dec.	α_{95}	K	VGP Latitude	VGP Longitude
Benguet	79-25	Rosario	N16.5	E120.5	5	100/—	22.2	348.2	7.3	110	N77.6	E9.4
	79-26	Rosario	N16.5	E120.5	6	300/380	−9.4	213.1	9.8	62	S55.5	E46.0
	78-10	Rosario†	N16.5	E120.5	13	400/—	−40.0	198.0	4.5	86	S71.9	E7.3
	81-29	flow	N16.3	E120.5	4	—/425	12.3	9.3	11.9	60	N76.4	W102.5
Albay	81-202	flow	N13.5	E123.6	8	300/550	13.4	352.5	3.4	265	N80.1	W7.4
	81-204	flow	N13.5	E123.8	6	300/550	15.3	357.6	5.8	132	N84.1	W32.5
	81-205	flow	N13.5	E123.8	8	300/550	24.3	339.6	6.8	67	N70.1	E34.6
Marinduque	81-303	Gasan	N13.5	E122.0	9	150/—	21.6	0.2	9.1	38	N87.7	W62.9
	81-304	dike	N13.5	E122.0	5	200/—	16.1	6.6	5.1	212	N81.6	W109.5

N is the number of individually oriented paleomagnetic samples; Inc., inclination; Dec., declination; VGP, virtual geomagnetic pole.
*Demagnetization values are listed for first A.F. demagnetization in Oersteds, then thermal demagnetization in degrees Celsius.
†K-Ar age of 9.6 ± 0.5 m.y.

TABLE 3. Early Mid-Miocene Paleomagnetic Results

Province	Site	Formation	Location		N	Demagnetization*	Inc.	Dec.	α_{95}	K	VGP Latitude	VGP Longitude
Benguet	78-1	intrusion†	N16.4	E120.6	9	200/—	23.7	353.3	8.9	34	N82.4	E0.2
	79-4	Klondyke	N16.2	E120.5	6	150/—	27.4	350.9	7.6	78	N81.1	E20.3
	79-20	Klondyke	N16.4	E120.5	7	150/—	30.3	351.0	9.1	45	N81.2	E31.7
	79-21	Klondyke	N16.4	E120.5	6	150/—	36.8	346.6	8.9	57	N76.6	E50.9
	79-22	Klondyke	N16.4	E120.5	9	150/—	35.9	344.0	6.7	61	N74.4	E46.2
	79-30	Klondyke	N16.2	E120.5	5	100/—	26.6	336.6	13.1	35	N67.2	E28.0
	78-7	dike‡	N16.3	E120.6	8	200/500	−35.0	158.0	6.7	70	S68.9	W137.9
	79-9	zigzag	N16.3	E120.6	7	150/—	39.0	342.0	7.0	74	N72.1	E52.1
	79-11	zigzag	N16.3	E120.6	4	200/300	16.0	339.0	11.1	69	N67.9	E11.4
	79-28	zigzag	N16.3	E120.6	10	200/—	−28.0	160.7	5.3	85	S71.4	W151.2
Tarlac	79-14	Moriones	N15.5	E120.7	6	200/300	20.7	333.4	20.1	11	N63.6	E23.1
Marinduque	81-307	Mahinhin§	N13.3	E122.0	5	150/400	−28.6	141.2	17.7	20	S53.3	W140.4
	81-301	dike	N13.5	E122.0	5	—/300	−16.3	126.5	21.2	14	S37.3	W147.8

N is the number of individually oriented paleomagnetic samples; Inc., inclination; Dec., declination; VGP, virtual geomagnetic pole.
*Demagnetization values are listed for first A.F. demagnetization in Oersteds, then thermal demagnetization in degrees Celsius.
†K-Ar age of 12.4 ± 0.6 m.y.
‡K-Ar age of 15.0 ± 1.6 m.y.
§K-Ar ages of 17.5 and 15.0 m.y. [*Wolfe*, 1981].

TABLE 4. Oligocene Paleomagnetic Results

Province	Site	Formation	Location		N	Demagnetization*	Inc.	Dec.	α_{95}	K	VGP Latitude	VGP Longitude
Benguet	79-10	Pugo	N16.5	E120.5	4	200/—	26.8	339.2	5.5	284	N69.5	E27.3
	81-38	Pugo	N16.5	E120.5	6	300/—	−9.2	146.8	6.1	121	S53.0	W163.9
Vizcaya	81-40	flow	N16.5	E121.2	12	200/500	2.5	326.7	3.6	148	N15.8	E9.7
Zambales	79-7	sediments	N16.1	E120.1	4	300/550	9.0	81.5	5.0	334	N9.0	W151.9

N is the number of individually oriented paleomagnetic samples; Inc., inclination; Dec., declination; VGP, virtual geomagnetic pole.
*Demagnetization values are listed for first A.F. demagnetization in Oersteds, then thermal demagnetization in degrees Celsius.

TABLE 5. Eocene Paleomagnetic Results

Province	Site	Formation	Location		N	Demagnetization*	Inc.	Dec.	α_{95}	K	VGB Latitude	VGB Longitude
Tarlac	79-12	Aksitero	N15.5	E120.4	8	400/520	−0.9	288.2	10.9	27	N17.4	E26.7
	81-50	sills	N15.5	E120.3	10	200/550	31.3	108.1	7.1	47	S12.0	W171.3
Zambales	79-15	Ophiolite	N15.7	E120.1	10	400/—	17.2	98.8	7.2	46	S6.0	W160.8
	79-16	Ophiolite	N15.7	E120.1	6	400/—	−28.7	289.3	9.3	52	N13.6	E9.6
	79-18	Ophiolite	N15.7	E120.1	8	400/—	−12.3	115.6	6.5	73	S26.3	W150.7
	79-40	Ophiolite	N15.6	E120.0	12	400/—	1.3	120.0	6.3	49	S28.6	W159.6
	79-41	Ophiolite	N15.6	E120.0	5	300/—	−7.2	110.6	9.9	61	S20.8	W152.1
Pangasinan	79-42	stock	N16.1	E120.1	5	300/—	7.2	129.1	5.3	206	S35.9	W166.8
	81-43	sills	N16.1	E120.1	10	200/500	17.4	126.9	7.4	44	S31.9	W171.6
	81-45	gabbro	N16.1	E120.1	4	300/—	−11.8	125.5	12.8	53	S35.7	W154.2

N is the number of individually oriented paleomagnetic samples; Inc., inclination; Dec., declination; VGP, virtual geomagnetic pole.
*Demagnetization values are listed for first A.F. demagnetization in Oersteds, then thermal demagnetization in degrees Celsius.

Copyright American Geophysical Union

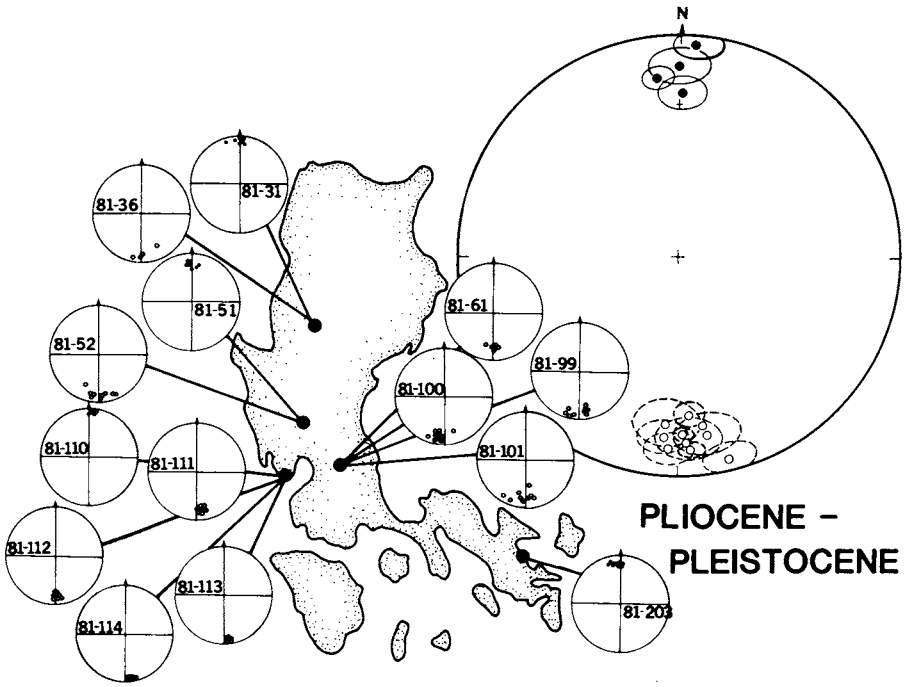


Fig. 6. Paleomagnetic results from sites in Luzon of Plio-Pleistocene age. All paleomagnetic directions are field directions (after tectonic correction), displayed on Lambert equal-area stereographic projections. The small stereo nets represent the paleomagnetic directions, after either thermal or alternating field demagnetization, for each sample from the indicated site. The solid circles show the approximate location of the sites. The large stereo net shows the site Fisher mean directions and their associated ovals of 95% confidence.

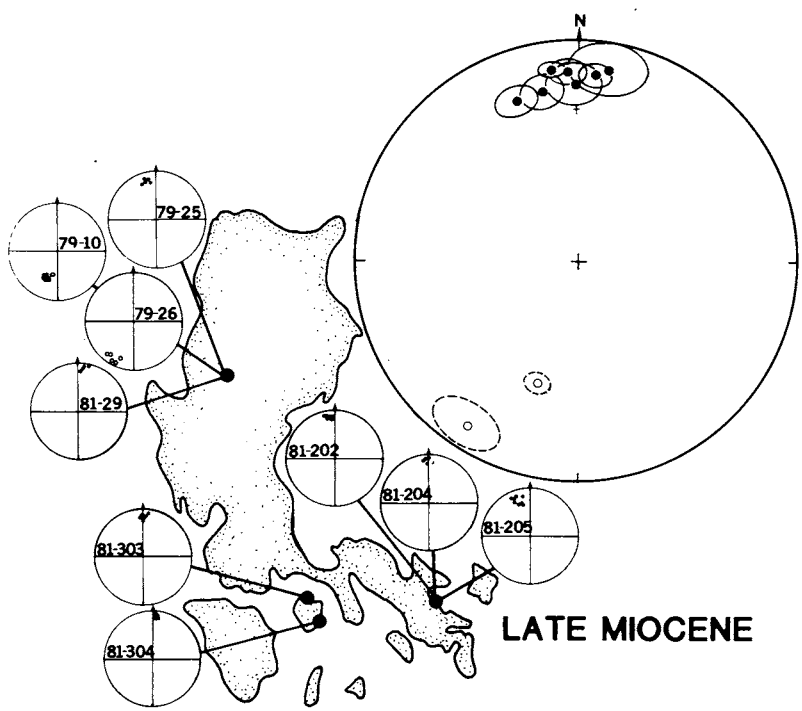


Figure 7. Paleomagnetic results for sites in Luzon and Marinduque of late Miocene age. (See legend for Figure 4.)

Copyright American Geophysical Union

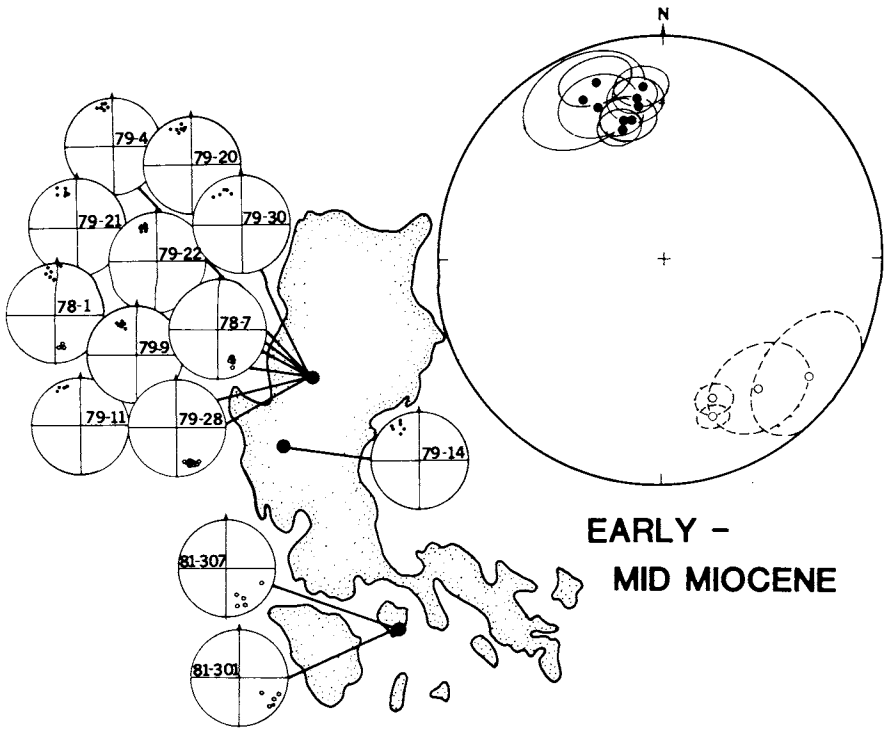


Figure 8. Paleomagnetic results from sites in Luzon and Marinduque of early to mid-Miocene age. (See legend for Figure 4.)

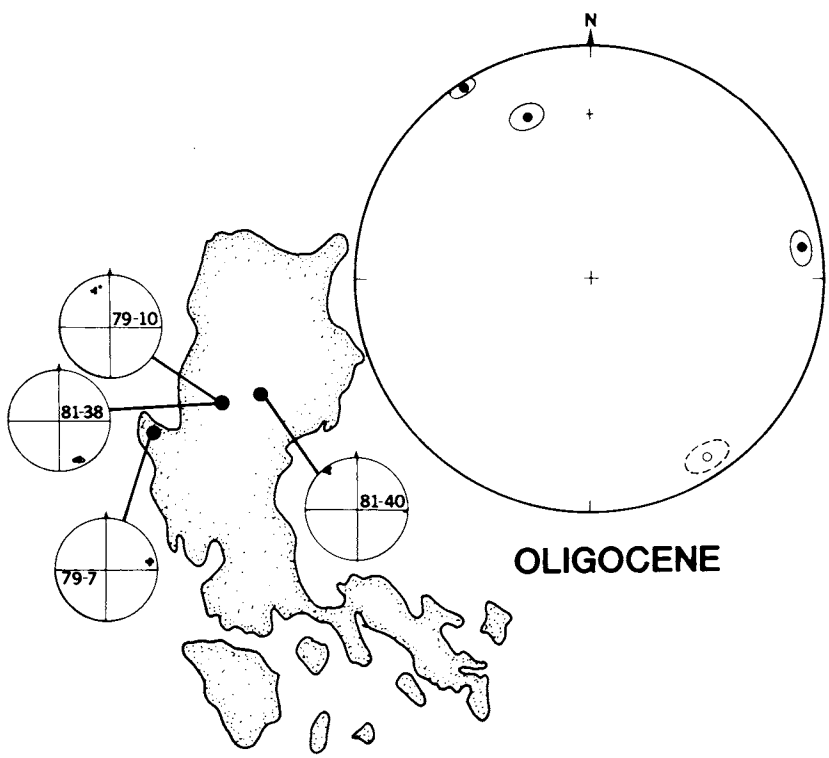


Figure 9. Paleomagnetic results from sites in Luzon of Oligocene age. (See legend for Figure 4.)

Copyright American Geophysical Union

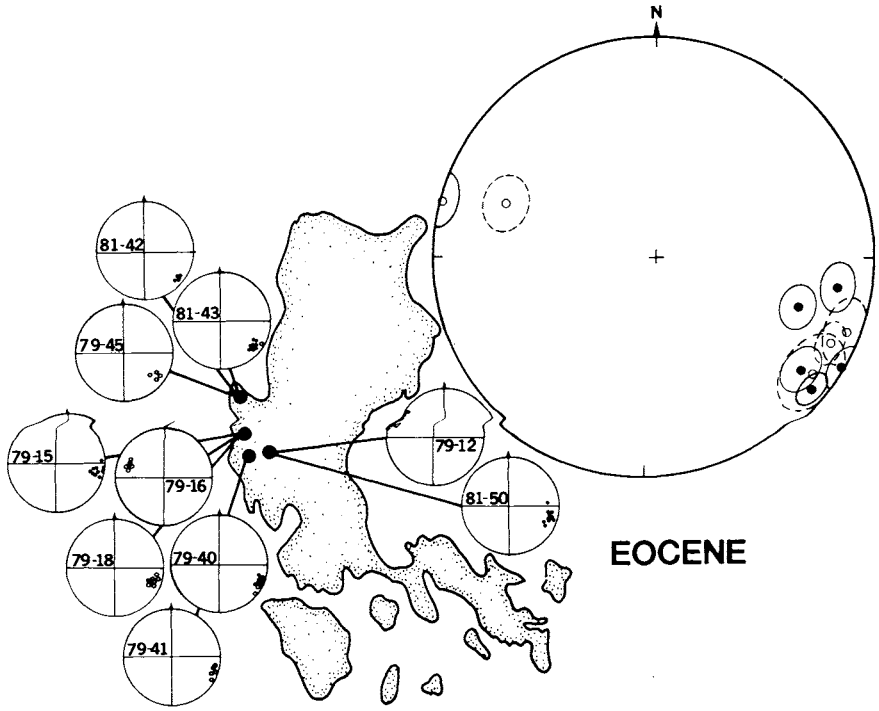


Figure 10. Paleomagnetic results from sites in Luzon of Eocene age. (See legend for Figure 4.)

are interpreted as sills [e.g., *Schweller and Karig,* 1982; *Hawkins and Evans,* this volume]. They (81-43) and the gabbros (81-45), into which they are intruded, are paleomagnetically good recorders, giving a southeasterly declination and shallow inclinations. A small stock (79-42), thought to be part of the ophiolite (J. W. Hawkins, personal communication, 1981) gives a similar direction before tectonic correction. A tectonic correction can be made by using the dip of the Oligocene sediments collected at Barlo; they lie conformably upon the pillows of the ophiolite there. The mean direction for the sills, the gabbro, and the diorite remains to the southeast as shown in Figure 10. A second possibility is to take the dip of the sills as an indicator of the horizontal at the time of acquisition of magnetization of these units. This gives a similar declination, but rotates the magnetization on to the upper hemisphere with an inclination of about $-20°$.

At the Acoje mine, samples from the gabbros, the olivine cumulate dunite, and from minor dykes gave self-consistent results before tectonic correction. The declinations were roughly 120 and 300, and the inclinations shallow with normal and reversed results approximately antipodal. Initially, we attempted to use the attitude of the layering in the gabbro as a tectonic correction. This, however, destroyed the antipodal relationship of the normal and reversed results. We then tried the same correction as had been used in the north, on the grounds that the Acoje Mine sites are part of northern or Acoje block of *Hawkins and Evans* [this volume]. This brought the results into agreement with those from the north, except the declinations were a little nearer to east-west. The mean inclination for these sites is 13°, using this tectonic correction. If, instead, we argue that the Aksitero sediments are closer to the Acoje Mine and use that correction, the declination is not changed much, but the inclination is made somewhat shallower, ~9°. The magnetization of these igneous units is carried by magnetite. The olivine cumulate is somewhat serpentized, so the origin of some of the magnetite in it may be associated with the serpentinization, but, in the gabbros it appears to be primary. The failure of the layering as a tectonic correction is probably because it is not a good indicator of horizontal. It may well be a reflection of the inclined walls of the magma chamber at the time of origin [*Irvine,* 1980; *Pallister and Hopson,* 1981]. We interpret this magnetization as a record of the field when the ophiolite was a part of the seafloor.

The southern, or Coto, block of the ophiolite was sampled on the eastern side of the Zambales range, where the upper reaches of the body can be seen with sediments on top. The Aksitero sediments, which lie on top, are particularly important because the sedimentary environments they record constrain tectonic models [*Garrison et al.,* 1979; *Schweller and Karig,* 1982]. Unfortunately, the Aksitero is not a very good paleomagnetic recorder; only the pelagic limestones at the base gave satisfactory results. Upsection, the results become inconsistent and confusing. We are

Copyright American Geophysical Union

making a detailed investigation of the magnetic carriers throughout the section to see if we can interpret a reliable direction from upsection. For the moment, we prefer to use only the results from the Eocene pelagic limestones in the bottom 5 m of the section. They gave a declination of 288 and essentially zero inclination. The intrusives in the Camelling River are interpreted as sills. Using the tectonic correction from the Aksitero, we determine a declination of 108 and an inclination of 31. If we use the sills as an indication of horizontal, the declination is rotated to the southeast, but the inclination is not strongly changed.

In interpreting the paleomagnetic results from units that were magnetized at low latitudes and subsequently subjected to large tectonic rotations, we encounter the all too familiar problem of polarity ambiguity. If we assume polarity, we know the sense of rotation. Alternatively, if we know the sense of rotation, we know the polarity. However, if we cannot safely demonstrate polarity, we do not know the sense of rotation. For Zambales, we can argue that the normal vector is to the west because the Miocene Moriones formation on top of the Aksitero has a declination of about 330° and a positive inclination. However, this interpretation could be contested on the grounds that the Moriones determination is not very good and because there is considerable time between the Miocene and the Eocene, in which large rotations could invalidate our interpretation.

In assessing the Eocene results, greater uncertainties have become apparent in the analysis than we encountered with the younger material. The results from the northern block give northwest-southeast declinations with shallow inclinations. At the Acoje Mine, the directions are more nearly east-west with shallow inclinations. The few results from the Coto block on the east side of the range are very similar to those from the Acoje Mine. Assuming that the Aksitero is a westerly, normal declination because of the Moriones result, all sites give a westerly normal declination.

4.6. *Summary and Discussion of Paleomagnetic Results*

The paleomagnetic results described above come from some 50 sites from the Cenozoic of the northern Philippines, so that although it is far from an exhaustive collection, it should give some indication of the regional paleomagnetic framework. The results are principally from Luzon.

The Plio-Pleistocene results give strong evidence of declination, over the past 5 million years, consistent with a geocentric axial dipole, although the inclinations are systematically shallow for reasons that we do not understand. It may of course be indicative of a strong g_2^0 component in the field, as has been suggested by others [e.g., *Merrill and McElhinny*, 1977]. In giving zero declination, our results differ from those of *De Boer et al.* [1980] who observed westerly declinations of as much as 35° in the past 4 m.y. These were interpreted by those authors as evidence of tectonic rotations. Although we did see individual flows in Bataan that had westerly normal declinations of as much as 15°, the mean result from Bataan was indistinguishable from zero. In other regions, there was no indication of nonzero declination.

The early and mid-Miocene, results give clear and unmistakable evidence of westerly declinations. The samples come from igneous and sedimentary units and include normal and reversed directions, and there are no particular difficulties in interpreting the tectonic corrections. We are therefore very confident that these early and mid-Miocene directions are reflecting westerly declinations of the geomagnetic field. The inclination is indistinguishable from the geocentric axial dipole field direction for the latitude of the Philippines.

In the light of the well-established early and mid-Miocene result, the late Miocene result is puzzling. The easterly declinations exhibited by the three sites from the Baguio region of Benguet are paleomagnetically reliable and should not be disregarded. They necessarily imply a very rapid change from the easterly declinations to the observed Plio-Pleistocene directions. The possibility of such easterly declinations introduced other complications into the analysis; secondary magnetizations of older rocks may be acquired not only with westerly declination, but also if they were acquired in the late Miocene could they have easterly declinations.

The Oligocene results from Baguio and Caraballc give westerly declinations comparable to those of the early mid-Miocene results. The results from the Barlo sediments are puzzling. Taken at face value, they suggest an eastern declination. However, the result depends upon the resolution of the polarity ambiguity, and the site is not totally reliable.

Given the resolution of the polarity ambiguity based upon the magnetization of the Miocene Moriones formation, the Eocene results all have westerly declinations and shallow inclinations. It must, however, be emphasized that these results are from units magnetized in low latitudes and subsequently strongly rotated. Hence, they are extremely difficult to interpret because of the polarity ambiguity. We prefer to interpret a westerly declination, but, until a lot more work is done, the result should be used with caution.

5. Tectonic Implications of the Paleomagnetic Results

The paleomagnetic results do not yet point clearly to any single model of the evolution of the Philippines. Indeed, in such a complex region, in which plate boundaries are changing radically on the time scale of the observations, it may not be possible to establish a unique history. Nevertheless, the paleomagnetic results do suggest certain shortcomings of some of the earlier tectonic models.

In this last section of the paper we note the constraints the paleomagnetic data can place on the tectonic models.

Copyright American Geophysical Union

Next, we see where discrepancies exist between the models and those constraints. Finally, we present, as a working hypothesis, the outline of a model of evolution consistent with the paleomagnetic data.

5.1. *Paleomagnetic Constraints on Tectonic Models of the Northern Philippines*

In making use of paleomagnetic data for tectonic reconstructions, a definition of discordance of a particular paleomagnetic result is necessary so that one can establish which tectonic units have been displaced. *Beck* [1980] has suggested a definition of discordance in terms of rotation and flattening. These quantities measure the departure of a particular paleomagnetic result from the direction proscribed by the appropriate reference pole. As Beck points out, this has the advantage of giving the discordance in terms of two parameters, which are readily related to tectonic rotations and N/S displacements. Unfortunately, in a region such as the Philippines, it is not at all clear what the appropriate reference pole should be. However, we assume that for at least the past 17 m.y. the Asian landmass to the north has behaved as a single tectonic unit. The paleomagnetic data for China for this period are not distinguishable from the geocentric axial dipole field (J. Lin, personal communication, 1981). For the past 17 m.y., we therefore use departures from this dipole as the measure of tectonic rotation and N/S displacements. For older periods, this would clearly not be justified and Lower Cretaceous rocks from China show declinations of about 20° east.

We suggest, from the paleomagnetic results the following tectonic constraints: (1) counterclockwise rotation of the region, predominantly since mid-Miocene time; (2) northward motion of the region, which ceased by early-Miocene time; (3) the northward motion of the Zambales complex from equatorial latitudes accompanied by counterclockwise rotation, since Eocene time; (4) the absence of paleomagnetically detectable rotations in the Plio-Pleistocene.

The value of such constraints clearly depends upon the reliability of the relevant paleomagnetic result. Thus, constraint 1 is the best established result, so any model should involve counterclockwise rotation since mid-Miocene. The northward motion of Benguet in the Oligocene is less well established. However, the results from the whole region from early-Miocene rocks strongly suggest that there was no important northward motion since that time. The northward motion of the Zambales complex and its counterclockwise rotation depend upon the resolution of the polarity ambiguity, but it is included as a weaker constraint. During the Miocene it became a part of Luzon and so subject to the constraint of no further northward motion. The Plio-Pleistocene results justify constraint 4. The possible clockwise rotation of the region north of the Philippine fault in northern Luzon has not been included as a constraint, because the paleomagnetic dates do not at present justify it.

In all use of paleomagnetic data for tectonic analysis, it should be remembered that determination of declination detects relatively small tectonic rotations, perhaps to the limit of geological interest. However, the resolution in the paleomagnetic determination of latitude, from inclination, is no better than ±500 km, even with very good data.

5.2. *Tectonic Models of the Philippines*

We now look at some of the earlier models in light of the paleomagnetic constraints established. It is then possible to see where discrepancies arise between models and the new data.

It is hard to reconcile with the paleomagnetic data models (such as that of *Murphy* [1973]) in which there is no plate boundary between Asia and the Philippines in the Cenozoic until middle Pliocene times. Our first constraint requires that sometime after mid-Miocene and prior to the Pliocene the region was rotating counterclockwise. Paleomagnetic results from the Asian landmass [e.g., *McElhinny,* 1973] suggest that it was not rotating in this sense. A plate boundary, or at least a collision zone, is then required between Asia and Luzon at this time; certainly, it must have been there before Pliocene times.

Karig [1973] proposed a model that invoked commencement of subduction, to the west of Luzon, in late Miocene time. This model is therefore more consistent with paleomagnetic data. It does not, of course, include the rotations that were reported after the model was proposed. The effect is to rotate the northern Philippines and the eastern subduction zone in Karig's model into a NE/SW orientation during the Early Tertiary.

Models in which the Philippines are rifted and rotated away from the Asian landmass, from China, in the Early Tertiary [e.g., *De Boer,* 1980] satisfy the counterclockwise rotation revealed by the Paleomagnetic work, but not the northward motion. They also encounter a fundamental difficulty in that spreading of the present South China Sea is younger than, for example, the Zambales ophiolite.

The documentation of the opening of the South China Sea by *Taylor and Hayes* [1980] has permitted more realistic models of the evolution of the region. They have shown that the South China Sea opened between anomaly 11 (32 m.y. or mid Oligocene) and anomaly 5D (17 or mid-Miocene). In a number of discussions that have followed, an essential role has been played by the southward motion of a continental block rifted from China [e.g., *Taylor and Hayes,* 1980; *Holloway,* 1982; *McCabe et al.,* 1982]. This continental block consisted of what is now the northern part of Palawan, Mindoro, Panay, and the Sulu Sea. Its importance is that it eventually collides with the subduction zone to the south of the South China Sea and stops subduction in this area. In the discussion of *McCabe et al.* [1982], the idea is taken further and used to account for

Copyright American Geophysical Union

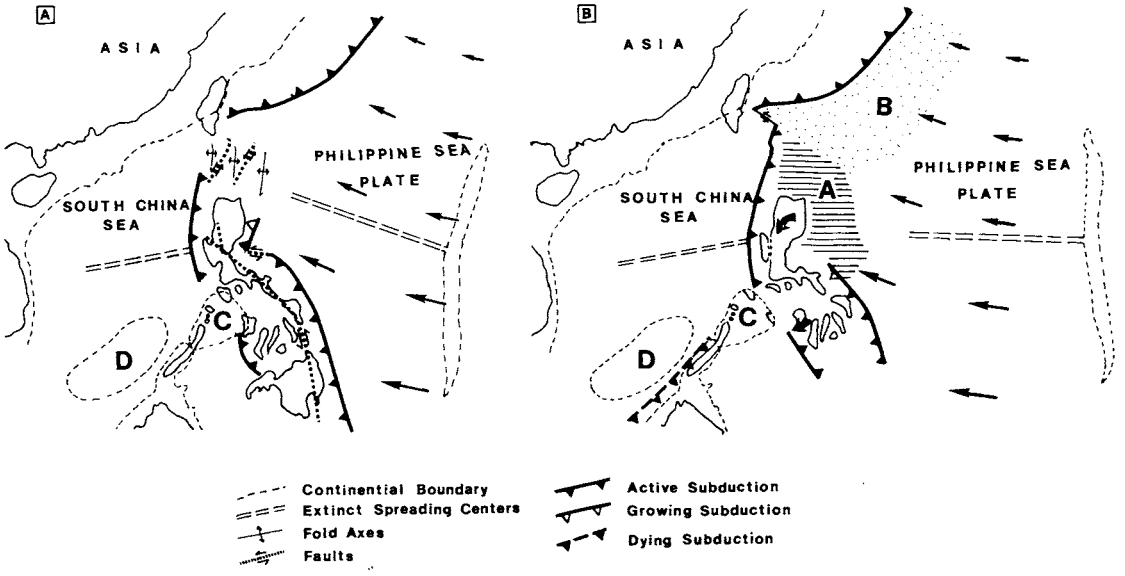


Figure 11. Tectonic maps for the South China Sea, Philippine Islands, and part of the Philippine Sea plate (*a*) at present and (*b*) proposed for approximately 17 m.y. ago (end of the mid-Miocene). The motion of the Philippine Sea plate is shown by a set of arrows, whose length approximates the velocity with respect to the Asian continent. Region A is Philippine Sea plate convergence during the time interval between the two maps. Region B is Philippine Sea plate that will be subducted by the Ryukyu trench during this same time interval. Region C is the Calamian continental fragment. Region D is the Reed bank continental fragment.

the difference in paleomagnetic directions found in Panay and the northern Philippines by invoking a collision between this block and the Philippines themselves. The collision of the block with the subduction zone is coincident with the cessation of spreading in the South China Sea. It appears that this collision indeed terminated spreading of the South China Sea.

5.3. *Paleomagnetically Based Tectonic Model of the Philippines*

The essence of a successful paleomagnetically based tectonic model is to define the motions of different tectonic units within the area of interest. This requires the identification of so-called terranes, whose boundaries are established. We cannot do this. However, we can give some indication of the motion of the region and begin to isolate these tectonic units. As a point of departure, we use the schematic tectonic map in Figure 11*a* based upon *Karig* [1982].

Our interpretation is that the Plio-Pleistocene results do not reveal any significant departures from the present configuration during the past 5 million years. Clockwise rotation of the late Miocene sites north of the Philippine fault in Luzon can be readily accommodated by rotating a unit whose southern boundary is the Philippine fault, but it is not required by the data.

The principal observation to be explained is the change between the situations shown in Figures 11*a* and 11*b*; the latter gives the tectonic configuration at the beginning of the middle Miocene (17 m.y.). It is immediately clear that we cannot explain this by invoking Philippine Sea plate motion directly, as it is in the opposite sense from that required. However, in an indirect manner this plate motion may cause the observed counterclockwise rotations.

At the cessation of spreading in the South China Sea, the Calamian continental fragment (labeled C in Figure 11*b*) collides with the trench at the southern boundary of the South China Sea and stops subduction. Therefore, the Philippine Sea plate cannot continue to move westward at this point. At this time there is evidence of volcanic activity on Samar [*Balce et al.*, 1981], which indicates the beginning of the development of the Philippine trench and subduction of the Philippine Sea plate under the central Philippines. This subduction takes up the relative motion between the Philippine Sea plate and Asia. Similarly, to the north, along the Ryukyu trench, the Philippine Sea plate is subducted (the portion of the plate labeled B in Figure 11*b*).

Between the growing Philippine trench and the Ryukyu trench, the leading edge of the Philippine Sea plate overrides the oceanic crust of the South China Sea. The Manila trench is effectively pinned by the collision of Calamian in the south, so that it is only free to migrate westward, toward Taiwan in the north. The extent of this mi-

Copyright American Geophysical Union

gration of the Manila trench is not sufficient to account for the amount of convergence of the Philippine Sea plate (labeled A in Figure 11*b*). Therefore, in conjunction with the subduction along the Manila trench, there is crustal shortening in the region north of Luzon, as indicated in Figure 11*a*. The westward migration of the northern section of the Manila trench, while the southern end is pinned by the Calamian continental block, results in a counterclockwise rotation of northern Luzon. Eventually, the northern end of the Manila trench collides and is pinned by Taiwan, so that counterclockwise rotation is terminated.

Prior to 17 m.y., the constraints from paleomagnetism are inadequate to justify an attempt at a model of tectonics comparable with that proposed for subsequent time. It appears that Zambales was formed at equatorial latitudes. It seems possible that the northward motion and associated counterclockwise rotation of Zambales may be related to the counterclockwise rotation of the north Borneo-Palawan spur of the Asian landmass [*Haile et al.*, 1977]. We have preliminary indications that the Cordillera of Luzon does not have a similar tectonic history to that of Zambales. Again, we have preliminary results from the Angat ophiolite, which suggest it was strongly clockwise rotated. It is now crucial to get paleomagnetic data to outline the various tectonic units, or terranes, and to see their rotations and displacements for this older time interval.

This study affords an example of how paleomagnetism can help in the interpretation of the tectonics of a complicated region. The paleomagnetic data focus on certain aspects of the geologic history, which must be explained by successful models of the evolution of the region. In this instance, the paleomagnetic data focus attention on the required counterclockwise rotation of the northern Philippines in the Miocene. To explain this, the combined action of the Philippine Sea plate motion and the pinning of the region in the south was recognized. It therefore appears that the paleomagnetic results have helped to establish the manner by which the Philippines is being accreted to the Asian landmass. The results therefore help to document the growth of Asia by the addition of products of island arcs.

Appendix: Sample Locality Descriptions

Pliocene-Pleistocene

81-31. Basaltic dike, which cuts through a hornblende andesite flow on Mt. Santo Tomas. Dike is not corrected for bedding, the intruded volcanics attitude is 305,18,S.

81-36. Basaltic dike that cuts vertically through the lower section of the Rosario Formation. Dike is located on Marcos Highway just west of city limit of Baguio 1 : 50,000 sheet. No bedding correction was used.

81-51; 81-52. Ash flow tuff. Two sites located just west of the McArthur Highway at kilometer 89 road post. Sites taken from two flow units. The site with the normal inclination is stratigraphically below the reversed site. These flows have small dip of less than 5 that is depositional.

81-61. Porphyritic plagioclase basalt flow with holocrystalline texture. Flow located 4 km from above Antipolo. Flow is horizontal.

81-99. Fine-grained volcanic ash fall deposit collected from water pipe trenches recently dug along the side of the geology building of the University of the Philippines campus. Ash deposits are horizontal.

81-100; 81-101. Fine-grained volcanic ash deposit (81-100) and ash fall deposit (81-101). Location is in a rock quarry located 15 km east of Cubao, Quezon City, on the northern side of the Marcos Highway. Layering of the ash is horizontal.

81-3. Hornblende andesite flow located along the east coast at a small fishing Barrio just east of Mt. Malingao. Dip is primary, not structural.

81-110; 81-111; 81-112; 81-113, 81-114. These five sites are located on the east side of the Mariveles Bay. Site 81-110 is located at the eastern edge of the harbor along the eastern boundary of the ship building pier. This unit makes up the basement on which the other four sites from this area sit. Bedding assumed horizontal. The other sites are from plagioclase hornblende dacite flows, which are exposed in three different quarries and along the sea cliffs. The bedding of these sites is horizontal, and the initial relationship of these samples with the erupting vents is easily seen in the field.

Early to Mid Miocene

78-1. Diorite pluton located on the east side of the Kennon Road at kilometer 238. Used homocline bedding of 320,20,W for tilt correction.

79-4. Well-bedded sandstone located on the east side of the Kennon Road 1 km of Camp 1. Attitude of bedding is 131,19,S.

79-20. Pyroclastic ash deposits located on the north side of the Asin Road, west of timber creek at a small north-south running stream. Attitude of layering deposits is 319,22,SW.

79-21. Fine-grained ash deposit located 1 km west of the Asin Road from site 79-20, along the Irisan Creek. Attitude of bedding is 318,29,SW.

79-22. Fine-grained tuffaceous sandstone located 7 km outer north side of the Asin Road near the Rosario-Klondyke content; bedding is 318,29,S.

79-30. Fine-grained sandstone located at kilometer 219 on the Kennon Road. Attitude of bedding is 147,30,SW.

78-7. Basaltic dike located at kilometer 241. Dike has been carved into large lion head. Used general Baguio homocline (320,20,W) to correct for tilt.

79-9. Fine-grained sandstone in Bued River along side of the Kennon Road, 1 km south of Camp 4. Attitude is 340,40,W.

79-11. Well-bedded green sandstone located in the river north of the houses located on the Kennon Road just below

Copyright American Geophysical Union

the dike carved in the form of a lion head. Attitude of beds is 0,25,W.

79-14. Calcareous sandstone located in the Moriones River near the town of Moriones, Tarlac. Attitude of bedding is 340,20,E.

79-28. Well-bedded red sandstone in the river 200 m south of site 79-11. Attitude is 336,63,W.

81-307. Hornblende quartz diorite. Samples taken from two mining pits, one belonging to Mar-Copper at drainage portal for south pit. The second site is located at Consolidated Mining, Inc., located around 30 km northwest of Mar-Copper. Samples from both locations were taken from barrier area of mines. Sediments above Mahinhin intrusions are horizontal, and, therefore, no bedding correction is used.

81-301. Basaltic dikes that intrude the Mar-Copper pits. Samples were collected from five dikes that proved to be nonmineralized. No bedding correction was applied because they intrude flat lying sediments about the Marinduque intrusions.

Late Miocene

79-25. Light-brown shale, which was collected around Barrio Calot on the Naguilian Road. Attitude of the bedding is 200,34,W.

79-26. Medium-to-coarse grained sandstone located on the Naguilian Road west of 3 km/site 79-25. Attitude of the bedding is 202,45,W.

78-10. Hornblende andesite flow in the Rosario Formation. Located 200 m west of 79-26 site on the Naguilian Road. Bedding correction for 79-26 is used.

81-29. Andesite flow on the road 200 m north of the road end at Mt. Santo Tomas. Attitude of interbedded ash on top of volcanic is 305,18,S.

81-202. Hornblende andesite flow located on the highway along the eastern slope of Mt. Malingao in the folded section of older volcanic basement of Mt. Isarog.

81-204. Hornblende andesite located southeast of Legaspi City, 10 km along the highway between Legaspi and Manila City. Attitude is horizontal, based on relationship with overlying ash.

81-205. Hornblende andesite, 3 km along the highway toward Manila from 81-204. Attitude of bedding is 355,42,W.

81-303. Fine-grained calcareous siltstone. Located at Barrio Balogo, across from the gate to the Mar-Copper Mining Company. Bedding is horizontal.

81-304. Dacitic dike that cuts vertically through Torrijos volcanics at Cagpoc Point, southeast of Marinduque. Sample not corrected for bedding of host volcanics. Attitude of Torrijos volcanics is 330,24,E.

Oligocene

B 79-10. Basalt flow located on the east side of the Kennon Road near kilometer 235 post. Attitude of upper surface is 153,32,W.

81-39. Andesite flow located around 10 km west of the airport on the west side of the road toward Philex Mines. Attitude at site is 270,55,N.

81-43. Red andesite flow on the east side of the road from Manila to Isabela at kilometer 212. Attitude dip of overlying ash bed is 72,28,N.

79-7. Tan colored tuffaceous siltstone situated above pillow section on the road to Barlo Mine. Attitude of siltstone is 284,13,N.

Eocene

79-12. Bottom 5 m of Aksitero section in type section of Aksitero river; bedding attitude is 320,20,E.

81-50. Sills in Camelling River to the west of Tarlac as shown on the map; bedding attitude of Aksitero used as tectonic correction.

79-15. Layered gabbro 700 m east of the contact with the Olivine cumulate at the Acoje Mine. Tectonic correction of following sites in ophiolite are discussed in text.

79-16. Olivine cumulate 1400 m west of the contact in the Acoje Mine.

79-18. Layered gabbro in stream section 15 km east of the Acoje Mine on the continuation of the Acoje Mine Road.

79-40. Layered gabbro on the Coto Access Road 30 km from Coto Mine.

79-41. Dyke that intrudes layered gabbro of site 79-40.

79-42. Stock near Sual on the coast road opposite Sauscito restaurant close to kilometer 349 post.

81-43. Sills on beach near Sual.

81-45. Gabbros intruded by sills of site 81-43.

Acknowledgments. The opportunity to carry out this study arose as part of the program in East Asian Tectonics and Resources of the United Nations Committee for Coordination of Joint Prospecting for Mineral Resources in Asian Offshore Areas. To this group and to the National Science Foundation we wish to express gratitude for financial support. Those of us from Santa Barbara wish to thank our many friends in the Philippines Bureau of Mines whose expertise was invaluable in the fieldwork, which was itself made possible by the logistic support of the bureau. We also thank K. Burton of Philippine Billiton Inc. and H. Pagauitan of Electro-Copper for many discussions and assistance with the field part of the studies. R. McCabe thanks S. Sasajima and members of his group for assistance in measurements.

References

Balce, G. R., R. Y. Encina, A. Momongan, and E. Lara, Geology of the Baguio district and its implications on the tectonic development of the Luzon central cordillera, *Geol. Paleo. S.E. Asia, 21,* 265–287, 1980.

Balce, G. R., O. A. Crispin, C. M. Samaniego, and C. R. Miranda, Metallogenesis in the Philippines: Explanatory text for the CGMW metallogenic map of the Philippines, *Geol. Surv. Jpn, 261,* 125–148, 1981.

Copyright American Geophysical Union

Beck, M. E., Paleomagnetic record of plate-margin tectonic processes along the western edge of North America, *J. Geophys. Res., 85,* 7115–7131, 1980.

Becker, F. G., Brief memorandum on the geology of the Philippine Islands, 20th Annual Report, U.S. Geol. Surv., 1899.

Chase, C. G., Plate kinematics: The Americas, East Africa, and the rest of the world, *Earth Planet. Sci. Lett., 37,* 355–368, 1978.

Corby, C. W., Geology and oil possibilities of the Philippines, *Philippine Dept. Agr. Nat. Res. Tech. Bull., 21,* 1–363, 1951.

DeBoer, J., L. A. Odom, P. C. Ragland, F. G. Snider, and N. R. Tolford, The Bataan orogene: Eastward subduction, tectonic rotations, and volcanism in the Western Pacific (Philippines), *Tectonophysics, 67,* 251–282, 1980.

Fisher, B. A., Dispersion on a sphere, *Proc. R. Soc. London, Ser. A, 217,* 295–305, 1953.

Garrison, R. E., E. Espiritu, L. J. Horan, and L. E. Mack, Petrology, sedimentology and diagenesis of hemipelagic limestones and tuffaceous turbidites in the Aksitero Formation (Eocene-Oligocene), Central Luzon, Philippines, *U.S. Geol. Surv. Prof. Pap., 1112,* 1–16, 1979.

Gervasio, F. C., Age and nature of orogenesis of the Philippines, *Tectonophysics, 4,* 379–402, 1967.

Haile, N. S., M. W. McElhinny, and I. McDougall, Paleomagnetic data and radiometric ages from the Cretaceous of West Kalimantao (Borneo) and their significance in interpreting regional structure, *Geol. Soc. London Q. J., 133,* 133–144, 1977.

Hawkins, J. W., and C. A. Evans, Geology of the Zambales Range, Luzon, Philippine Islands: Ophiolite derived from an island arc–back arc basin pair, this volume.

Holloway, N. H., The stratigraphic and tectonic relationship of Reed bank, north Palawan and Mindoro to the Asian mainland and its significance in the evolution of the China Sea, *Geol. Soc. Malaysia Bull.,* in press, 1982.

Hsu, I., Magnetic properties of igneous rocks in the northern Philippines, Ph.D. Thesis, Washington Univ., St. Louis, Mo., 1972.

Irvine, T. N., Magmatic density currents and cumulus processes, *Am. J. Sci., 280-A,* 1–58, 1980.

Jarrard, R. D., and S. Sasajima, Paleomagnetic synthesis for southeast Asia: Constraints on plate motions, in *The Tectonic and Geologic Evolution of Southeast Asian Seas and Islands, Geophys. Monogr.,* vol. 23, edited by D. E. Hayes, pp. 293–316, AGU, Washington, D. C., 1980.

Karig, D. E. Origin and development of marginal basins in the western Pacific, *J. Geophys. Res., 76,* 2542–2561, 1971.

Karig, D. E., Plate convergence between the Philippines and the Ryukyu islands, *Mar. Geol., 14,* 153–168, 1973.

Karig, D. E., Accreted terranes in the northern part of the Philippine archipelago, in *Proceedings of the SEATAR Workshop on Luzon-Marianas Transert,* in press, 1982.

McCabe, R., J. Almasco, and W. Diegor, Geologic and paleomagnetic evidence for a possible Miocene collision in western Panay, central Philippines, *Geology, 10,* 325–329, 1982.

McElhinny, M. W., *Paleomagnetism and Plate Tectonics,* Cambridge University Press, New York, 1973.

Merrill, R. T., and M. W. McElhinny, Anomalies in the time-averaged paleomagnetic field and their implications for the lower mantle, *Rev. Geophys. Space Phys., 15,* 309–323, 1977.

Murphy, R. W., The Manila trench-West Taiwan foldbelt: A flipped subduction zone, *Geol. Soc. Malaysia Bull., 6,* 27–42, 1973.

Pallister, J. S., and C. A. Hopson, Samail ophiolite plutonic suite: Field relations, phase variations, cryptic variation and layering and a model of a spreading ridge magma chamber, *J. Geophys. Res., 86,* 2593–2644, 1981.

Pena, R., Brief geology of a portion of the Baguio mineral district, *J. Geol. Soc. Phil., 24,* 41–43, 1970.

Philippine Bureau of Mines, Geologic map of the Philippines, 1964.

Schweller, W. J., and D. E. Karig, Emplacement of the Zambales Ophiolite into the West Luzon margin, *Am. Assoc. Petrol. Geol. Mem.,* in press, 1982.

Taylor, B., and D. E. Hayes, The tectonic evolution of the South China Sea Basin, in *The Tectonic and Geologic Evolution of Southeast Asia Seas and Islands, Geophys. Monogr.,* vol. 23, edited by D. E. Hayes, pp. 89–104, AGU, Washington, D. C., 1980.

Wolfe, J. A., Philippine geochronology, *J. Geol. Soc. Phil., 35,* 1–30, 1981.

Zijderveld, J. D. A., A. C. demagnetization of rocks: Analysis of results, in *Methods in Paleomagnetism,* edited by D. W. Collinson, K. M. Creer, and S. K. Runcorn, pp. 254–286, Elsevier, New York, 1967.

Copyright American Geophysical Union